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AI for the People: Use Cases for Government

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AI for the People: Use Cases for Government

Artificial intelligence (AI) is *the* news topic of the 2020s. From ChatGPT writing essays to chatbots answering questions to algorithms identifying cancers, AI is impacting and often disrupting how we live our lives. AI is proposed or in place in every domain and touches every function. In government, AI is relevant for departments from education and transportation to national security and public safety. Within these organizations, AI can facilitate the hiring of new employees, providing interactive responses with constituents, planning workloads, identifying anomalies, and forecasting natural disasters, to name a few applications.

The potential for good is extensive. AI is credited with improving the quality, efficiency and equity of many government services. Examples include:

- Narita Airport in Japan uses advanced robots to enhance security though detection of anomalies about people, baggage, and equipment. The airport is also using facial recognition software to facilitate passenger movement throughout the airport while maintaining security protocols. Face Express identifies passengers via a photo image at their first touchpoint at the airport enabling them to pass through the rest of the airport without interruption.[1]
- The US Veterans Administration is using AI to synthesize veteran feedback on the agency's services to identify performance trends and issues for detailed analysis.
- The US Patent and Trademark Office is using AI to search for materials that help determine the current state of the art as it evaluates patent applications.[2]
- The Rwandan government worked with Babylon Health to create chatbots capable of assisting the triage process for patients calling the hospital. Upon hearing the callers' symptoms, the triage tool would provide recommendations for accessing care.[3]
- The US National Oceanic and Atmospheric Administration is using AI to predict the location of heat islands so proactive responses can be implemented to reduce the negative health effects of excessive heat. [4]
- The Spanish government has teamed up with IBM's Watson for the use of this system to address questions about value added taxes. Since the introduction of Watson, the number of email inquiries to the tax authority has declined by 80 percent as questions were addressed by Watson.[5]

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[1] Narita International Airport Corporation, *Digital Transformation*," Automation and Labor Efficiency," https://www.naa.jp/en/airnarita/automation.html

[2] "The Government Is Using AI to Better Serve the Public," United States, https://ai.gov/ai-use-cases/

[3] Arunima Sarkar et al., "Chatbots RESET Framework: Rwanda Artificial Intelligence (AI) Triage," World Economic Forum, March 31, 2022, https://www.weforum.org/reports/chatbots-reset-framework-rwanda-artificial-intelligence-ai-triage-pilot

[4] AI Use Case Inventory (link to download spreadsheet), "The Government Is Using AI to Better Serve the Public," https://ai.gov/ai-use-cases/

[5] Cristina García-Herrera Blanco, "The Use of Artificial Intelligence by Tax Administrations: A Matter of Principles," Inter-American Center of Tax Administrations, https://www.ciat.org/the-use-of-artificial-intelligence-by-tax-administrations-a-matter-of-principles/?lang=en

On the other side of the ledger, AI applications have been criticized for biased analysis often based on limited and nonrepresentative training datasets. One area of concern is the use of AI in the criminal justice system. Studies raise questions about biased data used to train the models. The nonprofit news organization ProPublica documented numerous cases where algorithms appear to favor whites over blacks in their assessments.[6] Autonomous vehicles were found to have a significantly greater ability to identify adults and light-skinned pedestrian than children and dark-skinned pedestrians in low-light situations.[7]

A very public example was Amazon's employee recruiting algorithm, which was biased against women as a result of the training data used. "Amazon's computer models were trained to vet applicants by observing patterns in resumes submitted to the company over a 10-year period. Most came from men, a reflection of male dominance across the tech industry. In effect, Amazon's system taught itself that male candidates were preferable. It penalized resumes that included the word "women's," as in "women's chess club captain." [8]

Beyond concerns about bias, the broad application of AI raises other issues including cybersecurity, data privacy, deepfakes, and job loss.

Listening to the media, you might conclude that "AI everything" is just around the corner. That is not the case. The US Census Bureau survey of AI use found that in 2023 only 4 percent of business were using AI to produce goods and services. The most extensive use was in the information sector, where the number was just under 14 percent.[9] There are several barriers to broad adoption of AI. The first is trust. Until AI demonstrates it is trustworthy and provides better decisions than humans, its adoption will be limited. The industry term for fictitious information is "hallucination." The term means the AI perceives something is not really present. While this sounds benign, it is not; making up information, which large language models (LLMs) are prone to do, does not lead to the confidence necessary for broad use. Progress is being made but made-up AI information still abounds.

A second barrier is resources to support and operate AI applications. The development of robust AI algorithm requires distinctive expertise, including computer and data science, math and statistics, machine and deep learning, as well as problem-solving

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[6] Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, "Machine Bias," ProPublica, May 23, 2016, https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

[7] "Driverless Cars Worse at Detecting Children and Darker-skinned Pedestrians Say Scientists," Kings College London, August 23, 2023, https://www.kcl.ac.uk/news/driverless-cars-worse-at-detecting-children-and-darker-skinned-pedestrians-say-scientists#:~:text=The%20researchers%20also%20found%20that,increased%20issues%20for%20nighttime%20driving.

[8] Jeffrey Dastin, "Insight: Amazon Scraps Secret AI Recruiting Tool that Showed Bias Against Women," Reuters, October 10, 2018, https://www.reuters.com/article/us-amazon-com-jobs-automation-insight-idUSKCN1MK08G/

[9] Cory Breaux and Emin Dinlersoz, "Only 3.8% of Businesses Use AI to Produce Goods and Services, Highest Use in Information Sector, United States Census Bureau, November 28, 2023, https://www.census.gov/library/stories/2023/11/businesses-use-ai.html

skills. There is a limited number of people with this expertise. This is especially true in government organizations, which often have difficulty competing with the private sector for these distinctive skills. A second resource constraint is money. The development of AI systems is expensive. An AI-powered chatbot that leverages natural language processing and offers individualized responses to queries can cost in the range of $50,000 to $100,000 or more.[10] Even if funds are available, supply constraints for hardware can limit AI adoption. One more barrier is the time and energy needed to integrate the AI program into operations. The requirements included systems integration, process changes, and developing employees with the necessary skills. Moreover, AI implementation requires extensive change management internally within the organization and its customers. Change, especially regarding new technology that is replacing human decision-making, is never easy.

Given the broad range of AI applications and the need to tap the good and avoid the bad, the selection and prioritization of the right use cases is essential. This paper provides a framework for identifying and sequencing AI initiatives. The framework is then applied to the administration of the judicial system to illustrate how it works in practice. The overarching philosophy of the approach is to fix existing performance gaps; longer-term, the focus can shift to making good performance better.

I. AI Basics

AI means different things to different people. The Oxford English Dictionary defines it as "the theory and development of computer systems able to perform tasks that normally require human intelligence, such as visual perception, speech recognition, decision-making, and translating between languages."[11] The operative words are "normally require human intelligence."

There are four core characteristics of AI. It can:

- Make or support decisions.
- Combine human attributes.
- Evaluate data and take actions.
- Provide feedback for continuous improvement.

Another way of idenitfying AI systems is that they meet three requirements. The first is intentionality—they make decisions as humans would. The second is intelligence—AI uses data to make meaningful decisions. The third is AI systems' adaptability—they learn and adjust to new information.

[10] "Chatbot Development Cost in 2024," Antino, March 3, 2024, https://www.antino.com/blog/chatbot-development-cost

[11] Definition of "Artificial Intelligence, Oxford English Dictionary, https://www.oed.com/dictionary/artificial-intelligence_n

Segmentation of AI systems is also helpful context. One differentiator is how intelligence is imparted into the system. AI models historically have learned through testing different approaches and refining the approach based on results. Pretrained models (the "P" in GPT) are trained on massive databases with the intension that they deliver the right answer based on the data. Another differentiator is whether the models are supervised or unsupervised. Supervised model results are reviewed by humans and graded to refine the performance of the AI program. This approach has the benefits of minimizing hallucinations and finding biases, but it is very time consuming. Unsupervised systems allow the program to self-learn based on using large amounts of data to calibrate a model and then take a subset of the data and test the results. Based on the results the model is refined; human intervention is unneeded. The downside is no human is looking at the results.

These characteristics and requirements are exemplified by autonomous vehicles (AVs). AVs use AI to replace a myriad of decisions that humans are particularly good at, such as when to accelerate, when to brake, setting travel speed, waiting for pedestrians in the crosswalk, and pulling to the side of the road to allow emergency vehicles to pass. Making these decisions requires the combination of many human skills, from vision and hearing to reading and problem-solving. Evaluation of the operating environment and taking response actions is a continual process when driving. The AV is able to do the same and it learns as it goes, improving performance just as a human driver does with more experience. The promise of AVs is that they should significantly reduce the number of deadly vehicle crashes, since the majority result from human error.

AVs also exemplify the challenges to substituting computer programs for human decisions. Consider driving and coming to an intersection with stop signs in all directions. When vehicles arrive at the same time, who goes first? There are rules of the road to answer this question. In Massachusetts, the answer is the first in is the first to proceed and if two vehicles arrive at the same time, the vehicle to the right goes first. Compare the rules with what actually happens. In Boston, the outcome is more a function of an informal negotiation from afar than based on the rules. The most aggressive driver goes first. How do you program an autonomous vehicle: with the rules of the road or what people really do?

II. Use Case Considerations

A starting point for identifying high-value use cases is understanding where AI has comparative advantage. AI is particularly strong in:

- **Searching, finding, reading, and summarizing:** At the macro level, the most popular generative Large Language Models (LLMs) draw from the entire internet to search for information. They are also effective in synthesizing vast amounts of data and turning it into useful information. At the micro level, AI can also search a 1,000-page legal brief and find every time a specific item such as "endangered" is mentioned, as well as an array of synonyms including "threatened," "imperiled,"

"jeopardized," and "risked." The user does not even need to specify the synonyms, the algorithm can do it automatically based on its training.

- **Writing prose:** Need a birthday message for a friend? Need an op-ed? Need a training manual? LLMs are your solution. Enter a few prompts and out it comes in seconds. Need to make it easier to understand? Prompt the program to write for an eighth-grader and it is produced in seconds. Not only can AI write, but it can also write in multiple languages.

- **Interactively answering questions:** We have all become a customer served by computer response systems. "For a complaint press #2," for example. These defined systems are effective if they address your specific concern. If not, the result is frustration. AI chatbots have the ability to allow you to ask (via voice or text) a question and get a custom response to your specific concern, in a matter of seconds. These systems also allow you to refine your ask based on the response. For example, you might have the following exchange with a school department chatbot:

    - You: "What documents do I need to register my child for school?
    - AI: Proof of address such as a utility bill.
    - You: "I don't pay the utilities?"
    - AI: You can also use a letter from the city government mailed to your name and address.

  The results are fast and responsive information, 24x7x365 for the constituent and lower cost for the school department. The interaction can happen in literally dozens of languages.

- **Pattern and anomaly recognition:** AI systems are highly effective at finding patterns in numerical data or text. Humans are also good at these activities, but AI avoids human biases. We tend to see what we expect to see and what we are asked to focus on. The video where you count the number of basketball passes between people but miss the gorilla walking through the group is a powerful example.[12] Beyond biases, AI systems can process much larger volumes of information. AI anomaly identification capabilities are helping to provide security at airports, identify traffic bottlenecks, and predict disease outbreaks. AI systems are also used to identify and predict malfunctions and faults such as weakness in infrastructure or impending failure of machinery.

- **Sensing emotions:** Reading the body language of others is a very human trait. Some do it well, others less well. AI has the ability to sense emotions based on training data. AI systems have been trained based on facial and voice recognition as well as text to identify human emotions, from happiness and sadness to anger and anxiety. At one end of the spectrum, the information can predict mental

[12] Daniel Simons, "Selective Attention Test." March 10, 2010. Video, 1:21. https://www.google.com/search?client=safari&rls=en&q=gorilla+walking+across+basketball+court&ie=UTF-8&oe=UTF-8#fpstate=ive&vhid=zephyr:0&vld=cid:46d895f4,vid:vJG698U2Mvo,st:0&vssid=atritem-

health concerns and provide proactive interventions. At the other end, it can be used to target ads.

- **Integrating multiple inputs and projecting outcomes:** Accurate forecasts are the foundation for many services, from predicting storms and managing traffic to operating the electric grid and predicting health outcomes for individual patients. Predicting electricity production is illustrative. The AI algorithm integrates historical production trends, operational efficiency, weather forecasts, and weather conditions at a very detailed level to estimate the amount of electricity that will be needed and the optimal portfolio for generation. The better the prediction, the more reliable the electric grid.[13]

Note that this list is indicative not exhaustive. New AI capabilities are increasing daily.

AI offers many opportunities, but the public sector manager faces several unique challenges that impact the use case selection. The first is that AI decisions have a **political overlay**. Operational objectives must be combined with political realities in identifying viable use cases. Using an AI hiring algorithm may create a risk of bias that is greater than a mayor is willing to accept. A corollary factor is general risk aversion. Media and public scrutiny may limit risk-taking. Neither the governor nor the cabinet secretary is willing to lose their job because of an AI implementation gone bad.

A second concern for public organizations is the **procurement process**. Many government entities have strict purchasing rules that may dictate the selection of the low-cost supplier. If the low-cost provider is also the best-qualified supplier, the outcome is great. However, this is not always the case. Public procurement regulations also can limit collaboration with potential suppliers. AI is new and ever evolving. Close collaboration between the government and the provider are needed to modify specifications as the AI product is being developed and delivered. This is often challenging in the public sphere.

Multiple customers is another complexity faced as government organization bring in AI. Who is the customer for a mental health AI product? Is the customer the recipient of the services? The person's family? Civil society? Each has their own interests. Determining how to simultaneously satisfy all needs can be so daunting that the status quo persists. Finally, there are **endless opportunities**. The range of AI uses cases in government is broad and dense. This, combined with limited funds and expertise, elevates the need to carefully prioritize AI use cases. The next section provides a framework for identifying and prioritizing AI use cases.

[13] John Villali, "Advanced AI-Powered Energy Forecasting," IDC Research, Inc., July 2023, https://www.sas.com/content/dam/SAS/documents/analyst-reports-papers/en/idc-advanced-ai-powered-energy-forecasting-113587.pdf

III. Use Case Identification and Prioritization Framework

The foregoing materials highlight the importance of having a structured process to both identify and prioritize AI use cases. The goal of the process is to emerge with a list of AI applications that will enhance the delivery of government services, especially those where current performance is inadequate. The list should also reflect the ability of the organization to successfully implement the AI use case at an appropriate level of risk.

The framework consists of the following five steps:

- Step 1: Profile core functions (A specific problem-centered approach is described in Appendix A.)
- Step 2: Prioritize “reds”
- Step 3: Identify AI applications for top-priority reds
- Step 4: Assess AI solutions
- Step 5: Prioritize AI solutions

Each step is detailed below.

Step 1: Profile Core Functions

The starting point for identifying potential AI use cases is profiling the functions performed by the agency and highlighting those functions that are both critical and for which performance is not fully satisfactory. External and internal functions are included. The external functions are those providing service to constituents/customers. In the case of the registry of motor vehicles, for example, those functions include conducting driving permit exams and road tests, providing driver’s licenses, transferring ownership titles, registering vehicles, certifying insurance coverage, and driver safety awareness, to name a few. The internal functions range from hiring and developing staff to designing and implementing IT and operating service centers. Strive for a comprehensive list.

Next, map the functions based on criticality. A simple test of criticality is: Will managers be fired if the function is not readily available. This outcome is unlikely if a public safety campaign is delayed a few weeks, but likely if licenses cannot be issued for a few days. A more comprehensive approach involves four tasks. The first is to map stakeholder expectations. Who are the stakeholders and what do they want and expect and how do these align by function? The second task is to assess the power and interest of the stakeholders, again by function. Power includes political influence, funding, expertise, and legitimacy. Interest is how much of a priority this is for the stakeholder. The third task is to map the criticality of the agency functions on a spectrum from critical to less important, based on the stakeholder assessment. The final task is to consider the error of margin associated with the task. Is the error tolerance associated with AI acceptable? Interestingly, in the case of AVs, human error (i.e. driver caused crashes) is more acceptable than AI failures.

Step 2: Prioritize the "Reds"

Because resources are limited, the focus of adopting AI should be prioritized to the critical functions where there is significant room for improvement. A green/yellow/red performance assessment provides an easy to implement and communicate approach. Where possible, the assessment should be quantitative, such as wait times, processing times and/or customer satisfaction surveys. Qualitative evidence is also helpful, such as the number of positive versus negative ratings on Yelp-like sites. Manager observations and employee input also play a role in the assessment. The green/yellow/red designation is likely subjective, but having some measurements that define each is preferable where possible. The AI use case priorities are the most critical "reds."

For many government services, different constituents/customers have different needs and expectations. Segmenting users can tease out nuances. For example, in the registry example, drivers who can renew their license online view wait times differently from those who need to complete the renewal in person. Geography can be another important way to segment. Those who live far from the registry are likely to have little tolerance for failing to complete their transaction in a single trip with minimal waiting time. Equity is also a consideration. Having multilingual staff at some locations is more important than at others.

Step 3: Identify AI Applications for Top Priority Reds

AI is unlikely to be the solution to all the reds. There are functions where AI has a comparative advantage; yet others where non-AI solutions are preferable. Also, there are applications where AI has a well-established track record and others where it is not ready for prime time. The risks of adopting AI can be reduced by being a fast follower rather than a first mover. Regulation is also a factor. In some geographies, higher risk applications require extra scrutiny.

Another risk mitigation strategy is segmenting the reds based on back-office versus customer/constituent-facing activities. Back-office activities are often lower risk. Also, less-critical functions may be a good starting point.[14] The counterpoint is that AI is a major undertaking consuming funds, time, and expertise. Therefore, the use cases addressed need to generate enough value to ensure there is a strong positive return on investment.

Back office or constituent facing the applications require data for training and testing. An additional screen is the availability of that data. Today that data might not be collected and ready for use. Thinking about the long term, beginning to collect the data now might enable the effective use of AI in the future as new algorithms are developed.

A good starting point for identifying AI applications is seeing where AI is currently in use in the public sector. Beginning with existing applications ensures the use case works

[14] Insight from Darryl Slabe, HKS Master in Public Admisntration candidate.

and provides social proof helpful in selling the idea to internal and external stakeholders. Moreover, the people running the existing applications can be a source of insight.

Focusing on existing public sector applications ensures that the AI use is appropriate in the unique government context. The US federal government maintains an inventory of AI applications.[15] The list includes uses by 27 federal departments and agencies, from Justice and State to the Environmental Protection Agency and the National Science Foundation. For example, the Department of Homeland Security's AI inventory includes applications for headquarters functions, as well as uses in the subagencies including the Transportation Security Administration (TSA), Federal Emergency Management Agency (FEMA), Customs and Border Protection (CBP) and the Cybersecurity and Infrastructure Security Agency (CISA). Use case examples include:

- Geospatial damage assessments (FEMA)
- AI for Autonomous Situational Awareness (CBP)
- Automated Indicator Sharing – Scoring and Feedback (CISA)
- Touchless Precheck Identity Solution (TSA)

Scanning the hundreds of current AI use cases serves as an effective thought-starter for addressing the "reds."

Another source of AI applications in the public sector is the Government AI Coalition initiated by the city of San Jose, California, in 2023. The mission of the coalition is to provide a vehicle for state and local government to have a voice in architecting the AI future.[16] The coalition includes representatives from 250 state, county, and local government organizations. The group's work is guided by the following goals:

- Use AI for social good,
- Ensure ethical, nondiscriminatory, and responsible AI governance,
- Promote vendor accountability,
- Improve government services, and
- Foster cross-agency collaboration and knowledge-sharing.

They offer a range of resources to support the development of AI tools, including use case templates and a registry of AI vendors.

Private sector AI applications offer another source of insight, especially where the application can be implemented in the public sector without extensive modification. AI use cases in the private sector cover the gamut of industries, from law and insurance to real estate, finance, and manufacturing. The use cases include:[17]

- Chatbots for customer service;
- Productivity tools from scheduling to note taking to performance assessment;
- Fraud detection and credit scoring in finance;
- Market analysis and predicting prices in purchasing;

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[15] "The Government Is Using AI to Better Serve the Public," United States, https://ai.gov/ai-use-cases/

[16] Government AI Coalition, City of San Jose, https://www.sanjoseca.gov/your-government/departments-offices/information-technology/ai-reviews-algorithm-register/govai-coalition

[17] MindTitan, "AI Use Cases," https://mindtitan.com/resources/industry-use-cases/

- Quality assurance, predictive maintenance, and process automation in manufacturing;
- Research and records management in law; and
- Call centers in many sectors.

A third source is AI providers. The AI provider community can be segmented into (1) solution providers; (2) AI-powered product providers; and (3) AI consultants/integration providers. The solution providers offer AI products that address specific functions and/or industries. OpenAI's ChatGPT and IBM's Watson are examples of products that offer solutions that cut across functions and industries. For example, Watson can be used to create chatbots, streamline coding, and find insights in datasets. AI-powered product providers like Apple (Siri) and Amazon (Alexa) use AI to support use cases including research, personal assistants, and managing the internet of things at home. A cadre of AI consulting firms offers services to help design and implement AI applications. Their comparative advantages are AI expertise and customization to the specific client needs.

Another source is open source projects were developers great AI applications and make them available to the public. They are free and often have an extensive user base that continually improves the application. Example span the gamut from productivity to coding to science and research. These offering might not look pretty but they can be an effective starting point.

While AI applications are not prevalent in all organizations, it is likely that someone is using AI to address your reds. Try using AI to find that needle in the haystack, but make sure you verify.

Step 4: Assess AI Solutions

Just because an AI solution exists does not mean it should be adopted. Like any initiative, AI applications must have a positive return on investment. The benefits must outweigh the costs. The benefits include improvements in the quality, efficiency, and equity of services provided to customers. Internal improvements in effectiveness and efficiency are also part of the calculus.

A range of costs needs to be considered, including the money to acquire, implement and maintain the system as well as the staff time to incorporate the AI application into existing operations. There are also the costs of having staff monitor the application especially when it is first implemented. Given the newness of the technology the human intervention might last a long time. New data also adds to costs as more information has the potential to improve performance but also requires time to retrain and retest.

Change in general, and with AI in particular, introduces risks. These include operational risk, the costs resulting from the failure to deliver required services, and reputational risks if the AI implementation fails to meet expectations. As mentioned above, these risks are greater in the public sector where failures are likely to be very visible. The media and political critics can easily amplify news of a failure, reinforcing the damage to

the organization's reputation. An additional consideration is the capacity for AI change. Do you have the expertise to bring in a new application? Also, what other initiatives are crowded out by the AI adoption?

The costs and benefits should be quantified as much as possible. Many of the costs are comparatively easy to quantify. The funds paid to vendors are clear. If payments take place over time, they can be discounted to get a present-value cost. The time needed for training is also easily estimated, with a drop in productivity during a transition period. Risk is harder to quantify. One approach is to look at the impact of other failures on an organization. For example, how was the budget of an organization reduced when a failure took place? The risk assessment also needs to reflect the strength of the AI provider—reliability, responsiveness, expertise, bandwidth, and long-term financial viability.

Benefits are often more difficult to quantify. What is the value of a more responsive employee who has been hired based on an AI algorithm, for example? The following is a segmentation and progression for quantifying benefits.

- **Segment 1: Direct Financial Gains** — If the AI application reduces staffing levels, the avoided labor costs are in the benefit set. Or if AI reduces raw material needs and space required, the eliminated expenses are included.
- **Segment 2**: **Improved Accuracy** — AI algorithms can often out perform humans. For example, AI programs often are better at detecting analogies in medical imaging than humans.
- **Segment 3: Customer Time Savings** — Returning to the registry of motor vehicles example, if AI scheduling reduces average wait time by 10 minutes per customer, the value of the time savings is estimated as a benefit. Time savings are routinely quantified for transportation infrastructure assessments. The cumulative time saved by all users is multiplied by an average wage rate to approximate the benefits.
- **Segment 4: Ancillary Benefits** — Some gains from AI adoption are indirect. For example, an AI traffic model that reduces congestion not only saves time but also reduces carbon emissions. Protocols for quantifying carbon reduction benefits are now well established.
- **Segment 5: Reputational Gains** — AI-improved performance can improve an organization's reputation, leading to a variety of benefits, including increased budget, enhanced recruiting of desired candidates, and less stress for employees.

The focus of the cost and benefit analysis should be on quantitative data, but qualitative information is also valuable. Descriptive narratives, especially of benefits, can often bring the benefits of AI to life. A customer who says they were able to get immediate and accurate help from a tax authority chatbot rather than wait on the phone for 20 minutes is powerful evidence of the value of AI.

There may be multiple AI options to address a specific red opportunity. In those cases, the benefit-cost analysis should be completed for each option. Choose the option with

the best return on investment. In making this determination, consider the working relationship you will establish with the AI provider. You will be partners for a long time, so make sure you have the right “chemistry.”

A final assessment is a make-buy decision. Should you develop and implement the AI application internally or partner with a third party? The assessment criteria include:

- Expertise
- Cost
- Time-to-use
- Customization
- Maintenance
- Data protection
- IP control

Expertise, cost and time are typically the threshold considerations. Where possible, quantify the assessment, but in most cases the final decision will be a qualitative.

Step 5: Prioritize AI Solutions

The prior step is completed for each of the top priority reds. Before launching initiatives that have a strong positive benefit-cost analysis, develop a road map to the overall AI plan. AI applications need to fit into the overall strategy and system of the organization. An AI application that, for example, provides more accurate forecasts of demand is necessary but not sufficient to improve customer satisfaction unless it is accompanied by a delivery system that is agile enough to use the improved forecasts.

Create a road map of how to sequence the AI applications that address immediate needs, the reds. Divide the map into a near-term plan (2–24 months) and a long-term view of 2–4 years. The sequencing should balance the comparative importance of the applications to performance with the risk-adjusted net benefits. Review and refine a draft mapping reflecting the organization’s overall strategy.

This task is competed by identifying one or two initiatives to adopt as a pilot, taking AI for a test drive in a low-stakes environment to learn what works and what does not. Those applications selected to pilot should have a high probability of success in light of the capabilities of the technology, the provider, and the implementing team.

IV. Framework Applied to the Judicial Court System

This section shows how the framework could be applied to the administration of the judicial system, specifically the courts. The content below is only illustrative; it is not based on a specific judicial system. The materials are intended to provide a general template and thought-starters for how to apply the framework in your own specific context. Citations are provided when factual information is used.

The core process stages of the court system are fourfold: pleadings, discovery, trial, and judgement. The completion of this process involves administrative tasks and substantive decisions. The five-step AI use case framework is applied below.

Step 1: Profile and Assess Core Functions

The starting point for identifying and prioritizing AI use cases in the court system is detailing the core functions performed. From the administrative perspective, managing the workforce, court facilities and physical evidence, documents, and digital content are core. The key functions are detailed in Exhibit 1.

Exhibit 1: Core Administrative Functions*

Managing the Workforce
- Hiring
- Training and retention
- Scheduling

Operating the Courts
- Scheduling
- Maintenance
- Language translation
- Document management
- Assign cases

Managing Evidence-Documents-Digital Content
- Storage and retrieval protocol
- Storage operations
- Security assurance

* Content for illustration purposes only.

The substantive tasks often involve legal research and analysis. Decision-making is the primary substantive responsibility of the courts and covers a broad spectrum of activities, the most important of which are setting bail, determining fault, and determining sanctions; sentencing in criminal cases; and compensation/restitution in civil matters. (Exhibit 2)

Exhibit 2: Core Substantiative Functions*

Research
- E-discovery
- Seeing patterns in jurisprudence

Analysis
- Summarizing case facts
- Assessing fault and liability
- Writing boilerplate
- Managing cases

Decision-making
- Probation/bail
- Sentencing
- Rehabilitation programs

* Content for illustration purposes only.

The assessment criteria for the core functions are quality, efficiency, and equity. From an administrative viewpoint, quality is a clear, predictable and reliable process. Efficiency is achieved when all resources, people, facilities, equipment/technology are used with minimal waste. Equity refers to equal treatment of all participants in the process regardless of race, socioeconomic status, location, gender, language, etc. Using red, yellow, and green as an assessment grade for “poor,” “all right,” and “good,” the following is the evaluation of the administrative core functions in our hypothetical court system:

- Managing the workforce: Red
  - Quality: Unpredictable schedules, poor retention (20% annual turnover versus 12% in other government agencies).
  - Efficiency: Recruitment process takes 3 weeks longer than benchmark.
  - Equity: Failing to meet diversity objectives.
- Managing the court facilities: Red
  - Quality: Facilities need $25 million to be in a state of good repair, frequent HVAC problems, spotty internet.
  - Efficiency: Maintenance costs 15% above industry standard; utilization of courtrooms in only 55%, 15% below industry standards.
  - Equity: Unreliable translation services cause delays and associated costs; courts in lower socioeconomic locations in a worse physical condition.
- Managing documents and digital content: Red
  - Quality: Document files well organized and easily accessed, but limited use of digital archival technology.
  - Efficiency: Accelerated use of digital storage offers $1.5 million in document retention budget.

    - Equity: All courts have access to the same document management process and technology.

The bottom line is that there is a significant opportunity for improvement in the administration of this judicial system.

Turning to the substantive functions, quality is defined as making decisions based on comprehensive, unbiased, and apolitical legal scholarship. It also reflects the accuracy of judicial decision-making. For example, was Driver A or Driver B at fault in a traffic crash? Efficiency addresses the time it takes to complete legal proceedings. Indicators are the total elapsed time for the process and the time judges can work on legal decisions rather than on administration. Equity centers on decisions that are consistent regardless of race, gender, socioeconomic status, or age, and judicial biases are minimized.

The assessment reveals the research and analysis activities are yellow and decision-making is green. Details and supporting rationale are below.

- Research and analysis: Yellow
    - Quality: Several decisions were overturned based on weak research/analysis scholarship.
    - Efficiency: Information is readily and easily accessible to all parties at the court and remotely.
    - Equity: Few decisions are overturned based on bias/discrimination.
- Decision-making: Green
    - Quality: Few decisions are appealed, fewer are overturned.
    - Efficiency: Court backlog and total elapsed time are less than benchmarks; judges spend less than 15% of their time on administration compared to the 20% benchmark.
    - Equity: Academic study of court decisions reveals minimal instances of bias in decision-making.

Exhibit 3 provides a summary of the assessments. These assessments are the inputs to segmentation and prioritization, the next step.

Exhibit 3: Assessment of Key Functions*

<u>Administrative Functions</u>

- Managing the Workforce
- Operating the Courts
- Managing Evidence/Documents/Digital Content

<u>Substantive Functions</u>

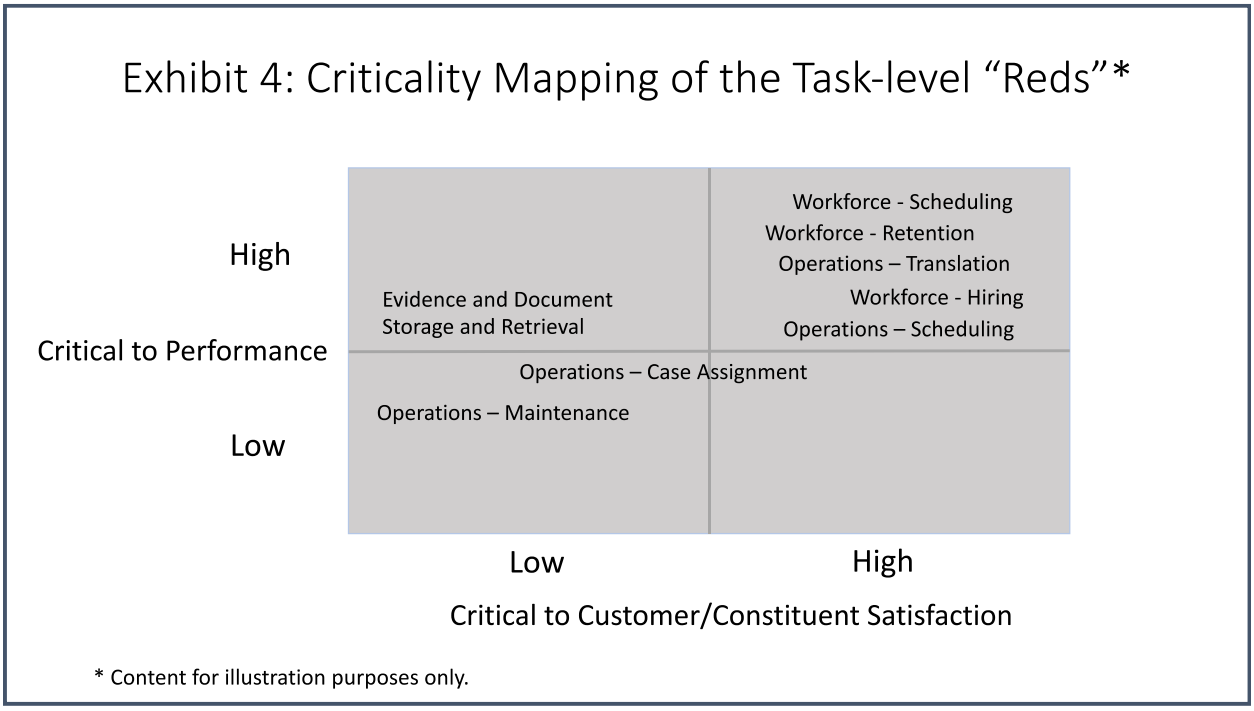
- Research
- Analysis
- Decision-Making

* Content for illustration purposes only.

Step 2: Segment and Prioritize the "Reds"

The potential applications of AI in most organization are expansive yet the capacity for adoption is limited. Therefore, prioritization is key. The functions profiled in Step 1 provided the reds, those activities where there is significant opportunity for improvement. In this illustration, these include managing the workforce, court facilities and evidence/document/digital content. These broad categories should be unpacked to specific activities, as AI applications are generally task specific. For example, the need for better management of the workforce is unpacked into hiring, training and retention, and scheduling.

The AI priorities emerge by considering the criticality of each specific task: critical to performance and critical to customers/constituents. Exhibit 4 shows the criticality mapping. Managing several aspects of the workforce, translation services and scheduling courtrooms are top priorities.



Step 3: Identify AI Applications for Top-priority Reds

The critical reds are the most pressing areas for improvement in the court system. Adopting AI is predicated on the availability of AI applications. As mentioned above, being a fast follower (using existing AI applications) is preferable to being on the leading (or often "bleeding") edge.

In this illustrative example, the one priority is providing translation services. The court system guarantees litigants, attorneys, and witnesses real-time translation to their native language for both oral and text communications. Routinely, translators are not available, leading to delays in proceedings which are costly for the court and its "customers." The root causes of this failure are (1) a lack of translators, especially in minority areas; (2) lack of translators in less widely used languages; (3) low compensation levels and no payment for travel time and cost; and (4) lack of sufficient lead time to arrange for services.

AI is routinely being used for translation in business and government. For example, US Customs and Border Protection developed CBP Translate to enabler its agents to communicate effectively with travelers who are not proficient in English when conventional translation services are not available. Protections are in place to obtain consent for the traveler to use the program; travel documents are not photographed;

biographical information is not captured for US citizens or Lawful Permanent Residents; and the conversation history is deleted after 90 days.[18]

Another example is the use of an AI translation chatbot by the Minnesota Department of Public Safety's Driver and Vehicle Services. The AI program provides translation in multiple languages including Spanish and Somali. The system provides interactive information such as updating insurance and updates in title and license status. The project was initiated as a result of customer input that indicated the language barrier was a major challenge for many in the state. The chatbot is powered by Google AI.[19]

In legal settings, however, the use of AI translation is still under consideration. Given the importance of addressing the court translation problem and that few options that exist, assessing AI translation is justified. Of course, the return on investment needs to be high to overcome the leading-edge risk.

This same process is completed for each of the red functions.

Step 4: Assess AI Solutions

Knowing that AI translation is used in government settings, the next task is determining if AI is a viable solution in your context and, indeed, the best solution. AI may be the best solution if the following conditions are met:

- AI providers have the capacity to design, implement and support your solution.
- Your organization has the expertise to use the AI application and management is prepared to support the change management process.

If these conditions are met, the next question is: Does using AI generate not only a positive return on investment but also a superior return compared to the non-AI alternatives? The return on investment is estimated based on the costs and benefits of the AI application. Exhibit 5 provides an illustrative example of the key factors for consideration.

[18] US Department of Homeland Security, "Privacy Impact Assessment for the CBP Translate Application," March 16, 2021, https://www.dhs.gov/sites/default/files/publications/privacy-pia-cbp069-cbptranslateapplication-march2021.pdf

[19] Julia Edinger, "Translation AI Helps Bridge Language Barrier for Minnesota DVS," *Government Technology*, September 28, 2023, https://www.govtech.com/artificial-intelligence/translation-ai-helps-bridge-language-barrier-for-minnesota-dvs

Exhibit 5: Cost and Benefit Factors*

Cost Factors

Direct
- Development or use fees
- Operating/computing and maintenance
- Quality control and security

Indirect
- Change management and training

Benefit Factors

Direct
- Faster proceedings
- Labor savings

Indirect
- More accurate and equitable outcomes
- Uniform translations

Risk Considerations
- Cost: Underestimate cost and time for implementation
- Benefit: Overestimate improvement in justice and savings

* Content for illustration purposes only.

The costs include:
- Initial development or use fee
- Ongoing operating/computing and maintenance
- Implementation, change management and training
- Quality control
- Privacy and security functions

Some of these are cash outlays for hardware, software, and expertise; others are using employee time and other organization resources. The costs are readily expressed in dollars. Of note, the cost of computing can be substantial. There is a tradeoff between more comprehensive models but their higher computing costs. Narrower models are likely to be more feasible for courts where budgets are likely constrained.

The benefits are internal efficiencies and external customer satisfaction. Internal efficiencies could include reduction in translation costs as AI substitutes for labor. The availability of AI translation at scale could also speed court proceedings, improving utilization of juridical resources. Moreover, AI is likely to enhance consistency through uniform translations. The labor reduction benefits can be directly converted into dollar savings. Resource efficiencies can also be estimated. For example, if proceeding delays waiting for translators are reduced by 10 minutes per day and there are 6 staff members impacted, there is an hour per day of labor savings. The value of this is the average labor cost per hour multiplied by the days per year the court operates.

The external customer benefits have both tangible and intangible dimensions. Tangible benefits center on faster proceedings. Less time waiting for translators is valued at an average wage rate. Likely of greater value is a better understanding of the proceedings

for the foreign language speaker and the other participants in the proceeding, which should lead to a more accurate and equitable outcome.

Valuing the customer benefits in general and the right outcome in particular is very difficult. One way to complete the cost-benefits analysis despite this challenge is conducting a breakeven analysis. Compare the costs with the directly quantified benefits. The difference represents the value that the indirect benefits must account for. If the number is small, it is likely the return on investment is positive.

Implementation of any new technology, including AI, has risks. The risks are on the cost side—higher than planned expenses and longer than anticipated implementation. There are also risks that the anticipated benefits will not materialize. Therefore, use conservative estimates and proceed only if the return is strongly positive.

A strong positive return on AI is necessary but not sufficient. The final requirement is that the return is higher than the return on non-AI alternatives. In the forgoing scenario, there were no other alternatives.

The bottom line in this example: If AI translation offers a strong positive return on investment, it should be on the list of AI applications for prioritization.

Step 5: Prioritize AI Solutions

The final step in the AI use case prioritization process is developing an overall AI road map. Two time frames are recommended: near-term (1–2 years) and long-term (3–5 years.) The near-term list of AI initiatives should focus on the current reds, where existing AI applications are ideally in use in the courts, but if not, at least in government. Use the “crawl-walk-run” approach to sequencing the projects. Do the easiest first to build internal capacity and minimize risk. Here the goal is to understand how best to bring AI into your organization. This also enables your organization to establish a track record of success and value creation, which will fuel subsequent initiatives. View the first few efforts as pilots, where learning is as important as generating the intended return on investment.

The long-term mapping reflects how the initiatives support the organization’s overarching strategy. Rather than addressing individual red functions, the long-term plan is directly aligned with mission and vision. A sample map for using AI in the courts is provided in Exhibit 6. Given the pace of innovation in the AI space, the long-term plan informs the indicative direction of the organization’s adoption of AI but should be revisited on a semi-annual basis to incorporate internal progress on AI use and new external applications being developed.

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Exhibit 6: Sample AI Priority Mapping*

Near-term Plan (next 24 months)	Long-term Plan (beyond 24 months)
Administrative Functions 1. Workforce – scheduling 2. Workforce – retention 3. Operations – translation 4. Workforce – hiring	Administrative Functions 1. Operations – scheduling 2. Evidence and document – storage and retrieval Substantive Functions 1. Research and information summary 2. Assessing fault and liability

* Content for illustration purposes only.

VI. Conclusion

AI is here to stay, and its impacts will continue to grow. Governments have an opportunity to leverage AI to delivery greater value to its constituents. The framework detailed above provides a structured approach to identifying and prioritizing AI use cases that maximize value and minimize risk.

Appendix A: Specific Problem-Centered Approach

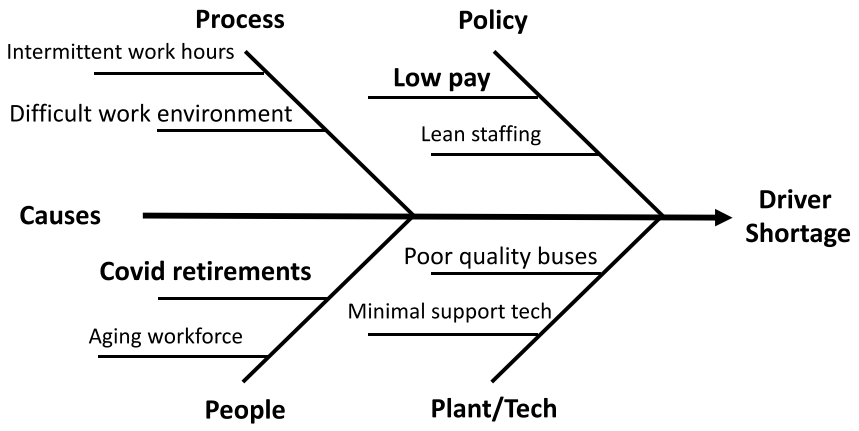
Steps 1 and 2 of the framework are designed to examine the entirety of the organization for AI use cases. However, some organizations may have a specific problem they need to tackle and seek to determine if AI offers a solution. In those cases, an alternative approach that narrows the focus of the investigation to addressing the problem is more efficient. This is accomplished by substituting the following tasks for Steps 1 and 2.

Task 1: Define the Problem and Assess Its Root Cause

Solving a problem begins with a clear articulation of the concern. The Kepner-Tregoe framework is effective in defining the problem. The framework asks the typical questions of what (identify the issue), where (locate the issue), when (timing of the concern and when did it first appear) and the magnitude of the issue. Importantly, it also asks the questions: When it is not a problem and where does it not appear? The answers offer important insights, enabling more specific problem definition.

A clear and crisp problem definition is necessary but not sufficient. The root cause of the issue must be determined. There are several approaches to getting to the root cause of the problem. An effective and intuitive approach is the Ishikawa or fishbone diagram. The head of the fish is the effect (i.e., the problem) and the bones are the causes. To ensure a comprehensive identification of the causes, ask: What is the role of polices, process, people, and plant/technology? These four dimensions form the core elements of the fish's skeletal structure. The smaller bones are the specific causes. When drawing the diagram, the more important the issue, the thicker the bone. An example of a fishbone root cause analysis is shown in Exhibit A-1..

Ishikawa Diagram of School Bus Shortage



Task 2: Identify Solution Options

There are often several options for solving a problem. Some might be policy or process related, others involve people. Others require a technology solution including AI. Several tools support the ideation process. One is SCAMPER. This acronym stands for substitute, combine, adapt, modify, put to another use, eliminate and reverse. These words provide a checklist to prompt your thinking on different ways the problem could be solved. Another tool is the ideation loop. You start by imagining a solution. Next you sketch what that solution looks like in reality. Then you create a model or test to see if it works. The idea of looking is that the process is iterative, continually refining and improving the solution.

Once you have a comprehensive list of potential solutions, you can evaluate them to find the most promising approach. Screening through the options is facilitated with a set of evaluation criteria. Efficacy, the direct improvement in the situation, is one key criterion. Others include operational, political, and financial feasibility. The criteria can be weighted to reflect their comparative importance.

The top option or two will become the focus for the AI assessment that begins in Step 3 of the AI use case framework presented in the main article.